PUNet: Printed Fabric Defect Segmentation Method Based on UNet

Jie Tang, Huanhuan Zhang, Zebin Su, Meng Jiang, Pengfei Li, and Junfeng Jing

***Abstract*— The contrast between printed defects and background texture is extremely low, and the defects are concealed within the patterns and colors, most existing detection methods are often unable to achieve promising results. This article proposes PUNet, a robust printed fabric defect network based on UNet. First, we designed the P1 block instead of the original convolutional layer of UNet. The P1 block applies a multiscale dilated convolution, enabling the model with stronger generalization and be more lightweight. Second, we introduced a polarized self-attention (PSA) mechanism, which enhances defect detail extraction and captures sensitive information about defective regions of printed fabrics, thereby improving the model's performance for small-sized defect segmentation. Extensive experimental results demonstrate that, compared to the original UNet, our proposed method significantly reduces parameters and computational costs by 63.7% and 58.5%, respectively, on the printed defect dataset and ceramic tile defect dataset. The mean intersection over union (mIOU) achieved is 80.66% and 85.75%, respectively, on the printed fabric defect dataset (PFDD) and tile surface defect dataset (TSDD). Furthermore, PUNet exhibits good detection performance on various representative industrial defect datasets. Compared to the selected state-of-the-art methods, PUNet outperforms them both qualitatively and quantitatively.**

***Index Terms*— Defect detection, multiscale features, semantic segmentation.**

I. Introduction

THE printed fabric, endowed with a high level of added value, commands a significant demand in various industrial sectors, including military equipment, clothing manufacturing, high-end home textiles, and interior decorations. Nevertheless, during the production of printed fabrics, various printing defects are unavoidable due to various factors such as the printing table, scraper, ink color, scraping operation, floral pattern movement, nozzle, and postprocessing. Fabric defects have a direct impact on the quality and pricing of the final product. Therefore, it is crucial to conduct defect detection to ensure their quality. Fabric defect detection is an indispensable step that can automatically identify and localize defects in the textile production process [1].

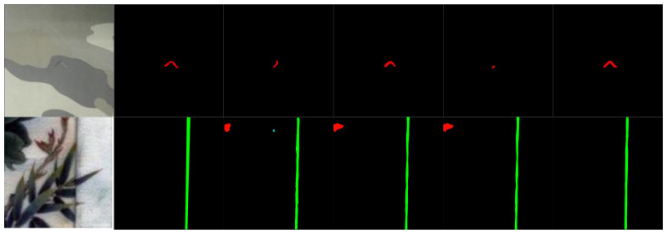


Fig. 1. Segmentation results using different methods of print defects. (a) Original images. (b) Ground truth. (c) UNet. (d) ResUNet. (e) UNet3+. (f) PUNet.

Unlike defects in other industrial products, the complexity and diversity of the background texture of printed fabrics make defect detection very difficult [2], [3], [4]. Due to the contrast between printed defects and background texture being extremely low, and the defects being concealed within the patterns and colors, most existing detection methods are often unable to achieve promising results, as shown in Fig. 1. Most of the surface defect detection methods [5], [6], [7], [8], [9] can achieve a good defect segmentation, but most of the methods are limited to simple or have a certain law of the background texture, it cannot deal with well for the background texture defects more complex printing fabric defects.

With the remarkable development of deep learning, most image segmentation methods can be effectively applied for defect detection, without requiring tailored designs [10]. In view of the success of UNet [50] and methods similar in structure to UNet (such as Res-UNet [11], FD-UNet [12], DC-UNet [13], MultiResUNet [14], KiU-Net [15], UNet-2022 [18], and DS-transUNet [19]) in medical images analysis and other fields, many industrial defect detection methods have tried to apply UNet and achieved excellent segmentation results. Jing et al. [27] proposed Mobile-UNet, which can achieve excellent segmentation results for small-size fabric defects, but a large number of depth separable convolution stacks will lead to the model in the loss of accuracy, and cannot be adapted to the complex texture background. Cheng et al. [20] proposed SC-UNet for segmenting fabric

Received 21 November 2024; accepted 8 January 2025. Date of publication 12 March 2025; date of current version 31 March 2025. This work was supported in part by the Key Research and Development Program of Shaanxi Province under Grant 2024GX-YBXM-231, in part by the Local Community Service Research Program Funded by Shaanxi Provincial Education Department under Program 24JC036, in part by the Natural Science Foundation of China under Grant 61902302, in part by the Preferential Funding for Postdoctoral Research Program in Zhejiang Province under Grant ZJ2022154, and in part by the Science and Technology Foundation of Xi'an for Program of University Science and Technology Scholar Serving Enterprise under Grant 22GXFW0033. The Associate Editor coordinating the review process was, Meng Jiang, Pengfei Li, and Dr. Dong Wang. *(Corresponding author: Huanhuan Zhang.)*

Jie Tang, Zebin Su, Meng Jiang, Pengfei Li, and Junfeng Jing are with the School of Electronic Information, Xi'an Polytechnic University, Xi'an 710048, China (e-mail: 3225033259@qq.com; szb505@126.com; m13259460281@163.com; li6208@163.com; 413066458@qq.com).

Huanhuan Zhang is with the School of Electronic Information, Xi'an Polytechnic University, Xi'an 710048, China, and also with the School of Automation, Northwestern Polytechnical University, Xi'an 710071, China (e-mail: zhanghuanhuan0557@163.com).

Digital Object Identifier 10.1109/TIM.2025.3550640

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defects and achieved excellent segmentation results, but it is limited to simple textured backgrounds.

Although these methods have achieved excellent segmentation results for fabric detection, most of them are limited to simple textured backgrounds and fail to solve complex real-world fabric defect detection problems. In our work, we propose PUNet, a robust printed fabric defect segmentation network based on UNet. The main contributions of our work are as follows.

1) We designed the P1 block instead of the original convolution module of UNet. The P1 block applied a multiscale dilated convolution, enabling the model with stronger generalization and be more lightweight. It exhibits superior segmentation performance for different complex texture backgrounds.
2) We introduced a polarized self-attention (PSA) mechanism, which can enhance defect detail and capture sensitive information about defective regions of printed fabrics, thereby improving the performance of the model for small-sized defect segmentation.
3) PUNet achieved a more stable segmentation effect in different backbones with lower computation cost and parameters, it has a more powerful generalization ability, and can effectively adapt to the complexity and diversity of the print defects.
4) Extensive experimental results demonstrate the superiority of our method to the selected state-of-the-art methods on various datasets in terms of both qualitatively and quantitatively.

The overall arrangement of this article is organized as follows. In Section II, we survey the existing literature that is relevant to our method. In Section III, we present the PUNet. In Section IV, we analyze the experimental results and compare PUNet with state-of-the-art methods. Finally, we conclude this article and give the future works.

II. Related Work

Traditional Defect Detection Methods: Most of the traditional defect detection methods utilize image analysis methods in the spatial or frequency domain to extract defect features. In spatial domain image representation, statistical methods [21], [22], [23] are used to extract image features for segmenting defects. Other approaches [7], [24] are to transform images from the spatial domain to the frequency domain and to analyze the regularity features exhibited by the image in the frequency domain to locate defects. Kang and Zhang [25] proposed a defect segmentation method that combines image integration with the Elo-rating algorithm based on the repetitive periodicity of fabric textures, effectively reducing the computation time. Liu et al. [7] developed a double sparse matrix to design an irregular background fabric model, it demonstrated strong performance in addressing different size (small, medium, and irregular) defects. Double sparse matrix effectively resolved the challenge of defect segmentation within complex background textures, but it is limited to a single background and defects, and cannot segment different fabric defects with varying printed backgrounds. Zhang et al. [26] proposed $L0$ gradient minimization to remove fabric textures, defects are segmented using the C-means clustering algorithm. Most traditional image detection methods are more effective for fabrics with relatively simple textures; however, the detection result is not good for printed fabrics with rich patterns, different shape defects, complex background textures, and low contrasts between defects and backgrounds.

Deep Learning Defect Detection Methods: Deep learning methods can address complex feature learning with stronger generalization ability, enabling them to adapt to various application scenarios. Existing defect detection methods can be roughly classified into supervised defect detection methods [10], [12], [27], [28], [29], semi-supervised and weakly supervised defect detection methods [30], [31], [32], [33], and unsupervised defect detection methods [34], [35], [36], which based on different training approaches. Wan et al. [9] proposed the locating, focusing, and refining network (LFR-Net) for saliency modeling. This model introduced edge cued through edge generating units (EGUs) and aggregated localization results, focusing results, and edge information into a high-quality detection map to enhance the detection performance of the model. Zhang and Ma [37] introduced a sensitive semantic segmentation approach (ClothNet) aimed at addressing the challenges associated with traditional semantic segmentation methods used for fabric defect detection. These challenges encompass the relatively high omission rate of small-size defects and the propensity for deeper network models to overlook the features of small-size defects, leading to subpar real-time performance. Based on the UNet network structure, Zhang et al. [34] subsequently proposed QA-USTNet and AGUR-Net [38]. Hu et al. [39] utilized a network structure that combines an encoder and deep convolutional generative adversarial network (DCGAN) to achieve fabric image reconstruction. Nevertheless, DCGAN did not perform well in detecting defects for print defect images with large resolutions and complex textures.

III. Method

In this section, we will present in detail the PUNet architecture, which consists of an encoder module and a decoder module, as illustrated in Fig. 2. The encoder module has P1 blocks and convolution layers, the decoder module has a PSA module (P2 block) and convolution layers. Details of the encoder module and decoder module are discussed in the following.

A. Encoder

The encoder module can extract features and obtain feature maps of different sizes for the channel. It consists of a convolution layer and a P1 block.

The design of the P1 block is inspired by concentrated-comprehensive convolution (C3) [40]. C3 is positioned in the residual connection from the input to the output, and it requires the input and output sizes and channels of the module to be consistent; thereby, it cannot be directly used for the downsampling processing. P1 block is composed of one 1×1 convolution and four dilated convolutions with

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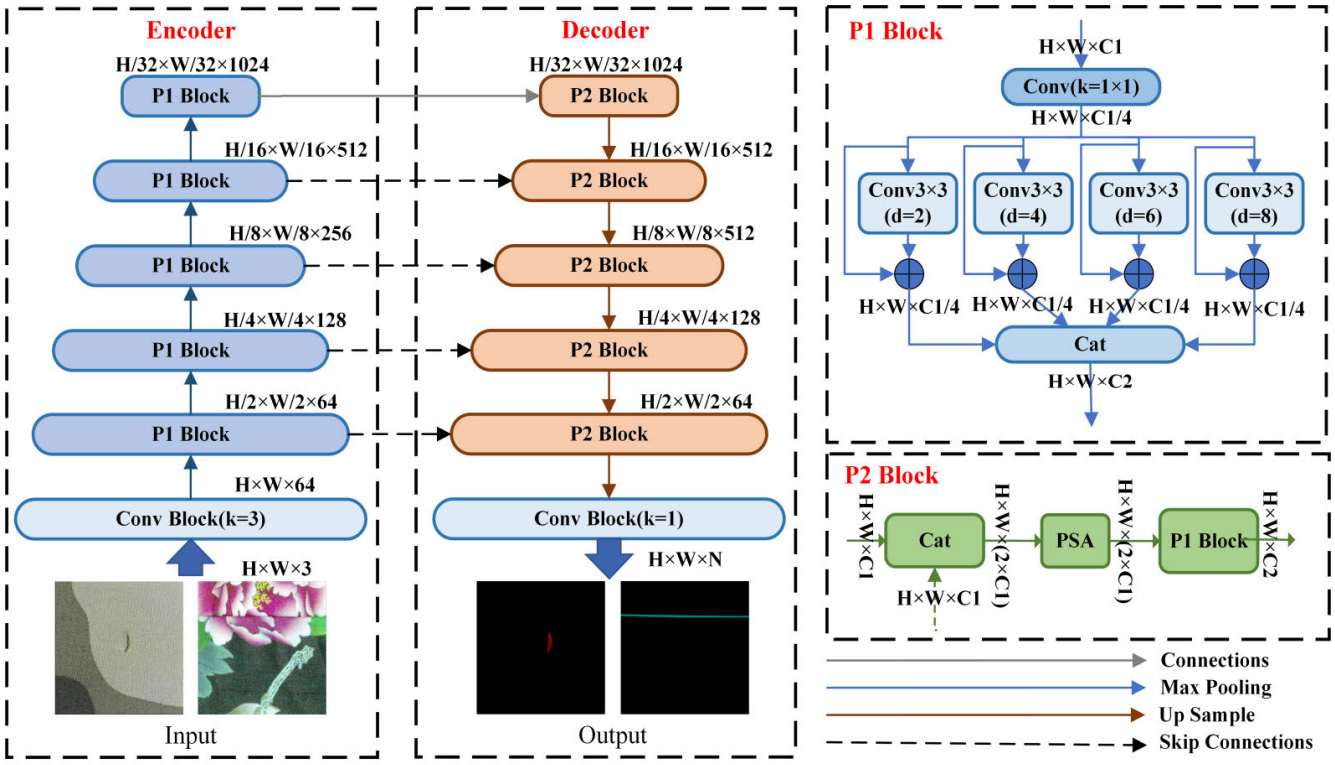


Fig. 2. Pipeline of PUNet and the individual module structures.

residual structures and dilatation rates of 2, 4, 6, and 8. The P1 block includes a 1 × 1 convolution within the module, reducing the number of parameters and computation cost. It also adds a residual connection for each dilated convolution branch, removing the original residual connection to prevent the loss.

The input of P1 block is denoted by x and the convolution is denoted by Conv_k. The first convolutional processing of the P1 block can be represented by the following equation:

$$y_{w\times h\times(c_1/4)} = \mathrm{Conv}\left(x_{w\times h\times c_1}\right) \tag{1}$$

where $w \times h$ denotes the feature map size, y denotes the output, and k denotes the convolution kernel size. The introduction of a 1 × 1 convolutional kernel size in the P1 block aims to reduce the number of parameters processed by each dilated convolution branch.

Equation (2) can be used to express the processing of the dilated convolution branch

$$a^i_{w\times h\times(c_2/4)} = \mathrm{DConv}^{d=i}_{k=3}(y) + y \quad (i = 2, 4, 6, 8) \tag{2}$$

$$\mathrm{Out}_{w\times h\times c_2} = \mathrm{Cat}\left(a^2_{w\times h\times \frac{c_2}{4}}, a^4_{w\times h\times \frac{c_2}{4}}, a^6_{w\times h\times \frac{c_2}{4}}, a^8_{w\times h\times \frac{c_2}{4}}\right). \tag{3}$$

The dilated convolution is denoted as DConv, where d is the dilation rate, c_1 and c_2 are the number of channels of the input and output, respectively. Based on the above representation, we can use Equation (3) to represent the complete processing of the P1 block, Cat is the channel splicing operation.

In the case of using the same size convolution kernel, the dilated convolution offers a larger receptive field compared to traditional convolution, which is advantageous for capturing subtle differences in the defective features of printed fabrics. The residual structure within each branch of the P1 block helps mitigate any offsets that may occur during the convolution process, preventing information loss from dilation, and enhancing the stability of the dilated convolution output. In addition, the multiscale approach enhances the robustness and generalization of the module. The convolution kernel at the input of the P1 block reduces the number of channels, thereby decreasing computational complexity and the number of processing parameters in each branch of the module.

Dilated convolution provides a wider receptive field, but the excessive stacking leads to certain positions in the feature map being overly emphasized, while the neighboring feature information is ignored, resulting in a discontinuous feature map (gridding effect)

$$a_i = \bigcap_i \mathrm{MSB}_{[2,4,6,8]}(x) \tag{4}$$

$$i_{t+1} = \bigcap_{i_{t+1}>i_t} (i_t \pm 4) \tag{5}$$

$$\mathrm{Out} = \mathrm{Cat}\left[a_{[2,6,4,8]}, a_{[4,8,2,6]}, a_{[6,2,8,4]}, a_{[8,4,6,2]}\right]. \tag{6}$$

The backbone of the PUNet model consists of P1 blocks, which are directly connected by maxpooling. We refer to the sawtooth design principle and incorporate different dilation rates in the P1 block to handle each multiscale branch (MSB) effectively. We use x to represent one of the MSBs in the first layer of the backbone. Then, the processing of the first layer of the MSB represents the result after processing the first layer of the MSB. The result of a single branch after undergoing processing through four layers can be represented by (4), where i represents the dilation rate for the first layer;

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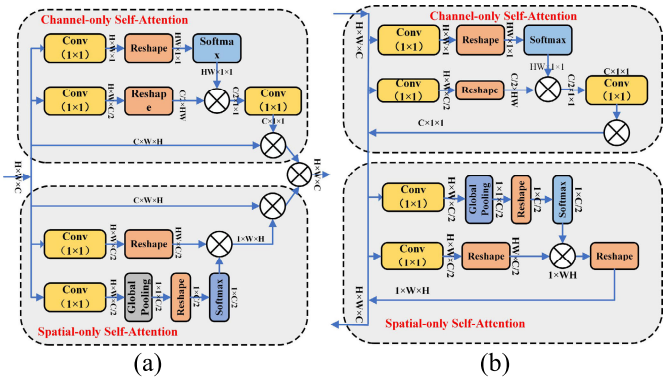


Fig. 3. PSA module. (a) Parallel structure of PSA. (b) Series structure of PSA.

$\cap$ represents the closing operation, meaning that the input of the previous layer is used as the output of the next layer; and a_i represents the output result of this branch. It notes that we set the dilation rates in a zigzag manner, with 2 and 6 forming one pair, and 3 and 8 forming another pair. For instance, if the dilation rate for the first layer of the MSB is 2, then the next layer's dilation rate will be set as 6, followed by 4 in the subsequent layer, and finally 8 in the last layer. These patterns continue to other branches as well, this process is illustrated in (5). The final output of the encoder after the aforementioned processing steps can be represented by (6).

B. Decoder

The decoder module consists of a P2 block and a convolution layer. The decoder is up-adopted from the feature maps of different sizes extracted by the encoder, the feature maps are extracted by using a linear layer and finally predicted for each pixel by the softmax function.

The P2 block is an integral part of the upsampling process, which consists of the P1 block and the PSA mechanism. PSA captures sensitive information about defective regions of printed fabrics, weighted in spatial and channel dimensions. P1 block captures global features and local features via different receptive fields. The combination of PSA and P1 block is crucial in the whole processing of the decoder.

The PSA consists of channel self-attention (CSA) and spatial self-attention (SSA). The overall structure of PSA has parallel and serial structures as shown in Fig. 3. The CSA can be expressed by the following equation:

$$y_1 = S[R[\mathrm{Conv}_{k=1}(x)]] \tag{7}$$

where x represents the input, Conv_k denotes a convolution with a kernel size of k, R represents the reshape, S is the softmax function, and y_1 is the output of the first branch; it transforms the size of $w \times h \times c$ the input into $hw \times 1 \times 1$ 2-D vector.

Similarly, the processing of the second branch can be represented by the following equation:

$$y_2 = \mathrm{Conv}_1(R[\mathrm{Conv}_1(x)] \times y_1) \tag{8}$$

where x denotes the input data or feature map that passes through the second branch, the size is transformed from $w \times h \times c$ to $c/2 \times h \times c$ a matrix of size. Multiplying the CSA vector size of $hw \times 1 \times 1$, we get vector size of $c/2 \times 1 \times 1$. Then, the vector undergoes an operation with a convolution kernel of size 1, followed by element-wise multiplication x with y_2. The overall expression of the CSA can be represented by the following equation, which yields a gain vector across channels via CSA:

$$y = y_2 \otimes x \tag{9}$$

where $\otimes$ represents the element-wise multiplication across each channel.

The structure of the spatial self-attention module can also be expressed using the same method, as shown in the following equation:

$$z = R(z_2) \otimes x \tag{10}$$

where z_2 represents a gain in the spatial dimension. The overall structure of PSA has parallel and serial structures, (11) represents the serial structure, and (12) represents the parallel structure

$$\mathrm{Out}_1 = (y_2 \otimes x) + (R(z_x \otimes x)) \tag{11}$$

$$\mathrm{Out}_2 = \left[R\left[z_2(y_2 \otimes x)\right]\right] \otimes \left[y_2 \otimes x\right]. \tag{12}$$

In the parallel structure, the input passes through both CSA and SSA. In the serial structure, the input passes through the CSA and the SSA in turn.

PUNet applied the P1 block to enhance the overall extraction and generalization ability of the network for defect information. However, the stacking of the P1 block throughout the process can make the feature maps different in channel dimensions, resulting in the sampling process of the encoder. PSA is introduced to enhance the spatial and channel dimensions of the feature maps after the P1 block and jump connection to suppress the effect of this difference. PSA is good at extracting detailed information on the object edges, enhancing the network recognition segmentation of defects in small-size printed fabrics, which is important for printing defect detection.

The P1 block enhances the generalization ability, the introduction of PSA reduces the difference caused by the P1 block and jump connection and enhances the encoder's extraction of the edge information of the details of the defects of printed fabrics, which makes the PUNet adaptable to the complexity and diversity of printing defects and achieves excellent segmentation results.

C. Loss Function

PUNet utilizes the Dice Loss function, as in (13), where Dice represents the Dice coefficient, $|X \cap Y|$ denotes the intersection between predictions and labels, and $|X| + |Y|$ is the sum of predicted and labeled elements

$$\mathrm{DiceLoss} = 1 - \mathrm{Dice} = 1 - \frac{2|X \cap Y|}{|X| + |Y|}. \tag{13}$$

For the binary classification problem, the pixels of the real segmentation labeling map are only 0 and 1, so (13) can be morphed into (14), where y_i and $\hat{y}_i$ denote the predicted and true values of the pixels, respectively. N denotes the total

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TABLE I
HARDWARE PARAMETERS AND CONFIGURATION

Name	Version/Size	Name	Version
GPU	Nvidia Geforce RTX 4090×2	Ubuntu	20.04
CPU	Intel Xeon(R) @24GHZ×32	Python	3.7
RAM	62.4GB	CUDA	11.1
VRAM	46.8GB	PyTorch	1.9.0

number of pixels. For the multiclassification problem, the loss function is calculated as shown in (15). n is the number of categories and DiceLoss_i denotes the DiceLoss on each category

$$\text{DiceLoss} = 1 - \text{Dice} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} \left(y_i + \hat{y}_i\right)} \tag{14}$$

$$\text{DiceLoss} = n - \sum_{i=1}^{n} \text{DiceLoss}_i. \tag{15}$$

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The detailed parameters of the experimental environment in this article regarding hardware configuration and experimental environment parameters are shown in Table I.

The methods trained and tested processes are all carried out under a unified framework. The Adam optimizer is used to optimize the training process, the initial learning rate is 1×10^{-4}, and the minimum learning rate is 1×10^{-6}.

The training set, testing set, and validation set of the dataset used for the experiments are all divided according to the ratio of 8:1:1, and the training set, testing set, and validation set used by the different methods in the same dataset are the same.

We utilize the evaluation metrics that are more commonly used in semantic segmentation to evaluate our method and other methods. In our experiments, we use pixel accuracy (PA), the mean PA (mPA), the mean intersection over union (mIOU), as well as the number of parameters (Params) and the amount of computation (Flops) to measure the quantitative segmentation performance of the different models on the dataset, whereas in the ablation experiments, we mainly use the category PA (CPA), intersection over union (IOU), to analysis the effect of each part of the PUNet on each type of the printed fabric defects, and some of the calculations are shown in the following equations:

$$\text{CPA} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{16}$$

$$\text{IOU} = \frac{X \cup Y}{X \cap Y} \tag{17}$$

$$\text{mIOU} = \frac{1}{N+1} \sum_{i=1}^{N} \frac{p_{ii}}{\sum_{i=0}^{k} p_{ij} + \sum_{i=0}^{k} p_{ji} - p_{ii}} \tag{18}$$

$$\text{mPA} = \frac{\sum_{i=1}^{N} \text{AP}_i}{N}. \tag{19}$$

TABLE II
CONFIGURATION STATISTICS RELATED TO DATASETS

Dataset		Class	Total	Train	Val	Test
PFDD		6	2519	2268	126	125
TSDD		5	1568	1255	157	155
MVTEC	Leather	5	368	295	48	25
	Pill	6	564	452	56	56
	Screw	5	476	381	48	47
	Wood	5	240	193	47	47
Crack500		2	1894	1516	186	192
ISIC2018		2	2694	2335	259	259

TABLE III
QUANTITATIVE ANALYSIS OF EXPERIMENTS COMPARING P1 BLOCK AND C3 MODULE

Method	PA(%)	mPA(%)	mIOU(%)
UNet	99.57	77.20	73.50
UNet+C3+Conv(k=1)	99.56	78.50	73.73
UNet+C3+Conv(k=3)	99.56	76.85	73.24
UNet+P1(PUNet)	**99.61**	**87.36**	**78.88**

We use PA, mPA, and mIOU to evaluate PUNet and different methods performance on printed fabric defect dataset (PFDD), tile surface defect dataset (TSDD) [41], and MVTEC dataset [42].

Since the ISIC2018 dataset [44] is a medical image dataset, we use the more common evaluation metrics for medical image segmentation, including sensitivity (SE), specificity (SP), accuracy (Acc), and IOU to evaluate the different methods and Crack500 is a dichotomous dataset, so we use the CPA to evaluate methods.

A. Datasets

We evaluate our method on five different datasets: PFDD, TSDD [41], MVTEC dataset [42], Crack500 dataset [43], and ISIC2018 dataset [44].

We provided a detailed explanation of the dataset used in the experimental section. The dataset division and related operations are outlined in Table II.

PFDD has six kinds of defects produced in the production of industrial fabrics, it has Hairy-Filaments (HF), Broken-Wefts (BF), Inks leak (IL), Stains (St), Wrinkles (Wr), and Pass (Pa) defects, the image acquisition is carried out by using the Basler acA2500-60uc area array camera, and the size of the images obtained is 1024 × 1024, total 1120 images. These images are resized to 512 × 512 and extended to 2560 images by rotating and panning.

TSDD [41] comprises five different defects on the surface of the tile, collected by Chinese Academy of Sciences in 2020. The dataset exhibits inconsistent image sizes, featuring a total of 397 defect images along with their annotations.

The MVTEC dataset [42] is a publicly accessible dataset designed for industrial defect detection. It was curated by the computer vision group at the University of Stuttgart, Germany. This dataset comprises high-resolution images of industrial products categorized into 15 different classes. Each

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TABLE IV
QUANTITATIVE ANALYSIS OF PUNET COMPARED TO OTHER METHODS ON PFDD AND TSDD

Dataset		PFDD			TSDD				
Method	Year	mPA(%)	mIOU(%)	PA(%)	mPA(%)	mIOU(%)	PA(%)	Flops(G)	Params(M)
UNet[50]	2015	77.20	73.50	99.64	78.42	66.43	96.61	46.15	31.04
DeeplabV3[45]	2018	77.72	73.61	99.60	86.63	67.57	97.89	6.61	5.81
ResUNet[11]	2018	84.59	76.16	**99.73**	86.84	82.65	98.64	22.94	43.93
UNet++[46]	2018	80.26	76.13	99.63	69.28	57.92	95.51	114.38	47.19
HRNet[52]	2019	84.14	79.09	99.61	87.82	81.02	98.56	4.64	9.64
U-Net3+[47]	2020	79.63	75.58	99.60	80.07	73.34	97.32	199.11	26.99
CMUNet[48]	2022	82.69	78.86	99.63	72.14	67.81	96.95	91.25	49.93
Mobile-UNet[27]	2022	**93.39**	52.28	54.22	68.13	32.35	37.88	13.11	6.05
CMUNeXt[49]	2023	81.53	79.32	99.64	82.32	68.12	94.45	**7.34**	**3.15**
PUNet	-	89.23	**80.66**	99.59	**92.15**	**85.75**	**98.72**	19.15	14.27

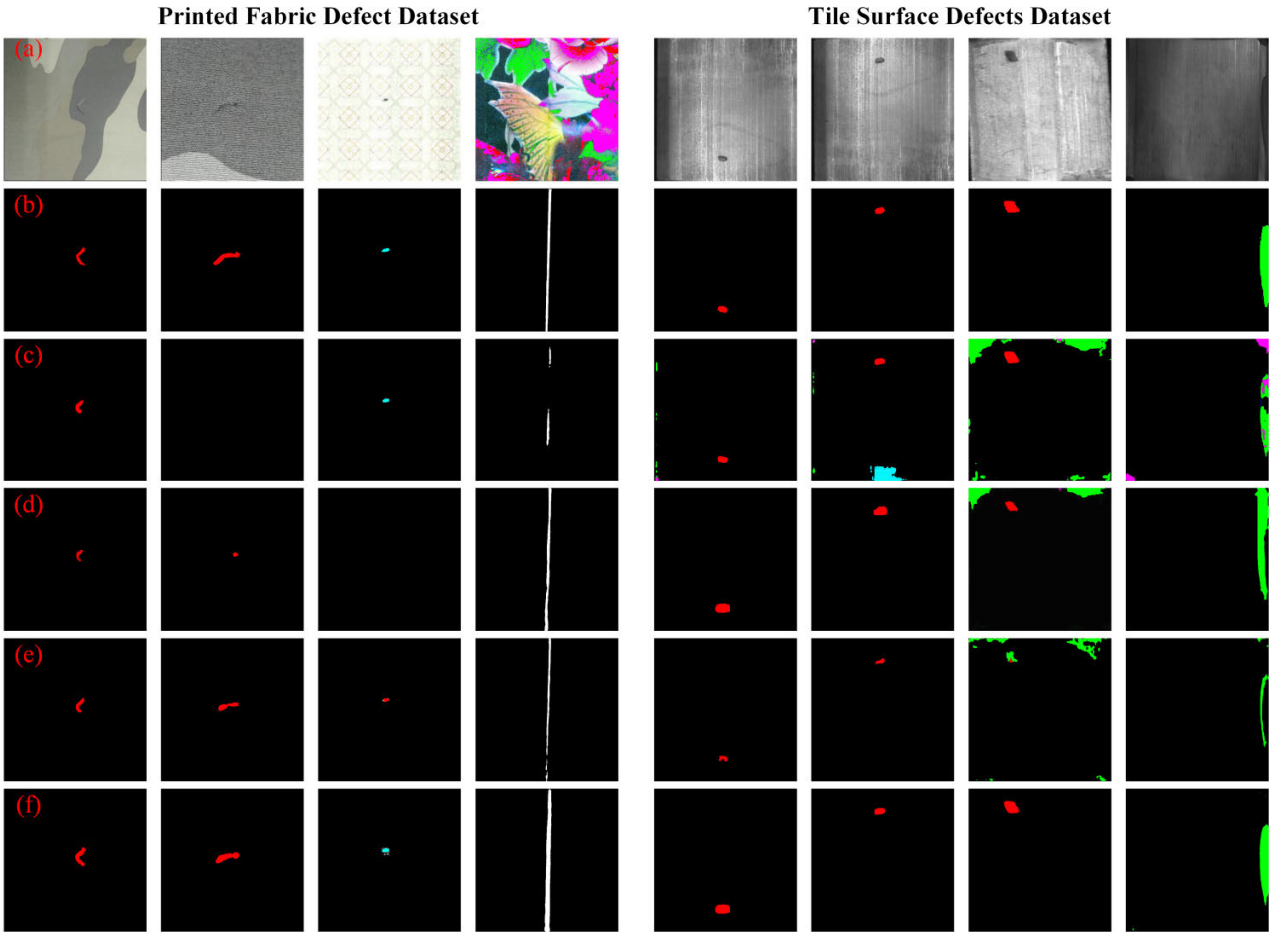


Fig. 4. Segmentation performance of PUNet and other methods. (a) Original images. (b) Labels. (c) UNet. (d) UNet3+. (e) CMUNet. (f) PUNet.

category includes both normal samples and various types of defects. We leverage several types of industrial defects from the MVTEC dataset that bear similarities to stamp defects.

The Crack500 dataset [43] contains 1896 images of road surface cracks and their corresponding labeling for various types and sizes of road cracks. We divide this dataset by organizing it into standard visual object classes (VOC) format for training.

The ISIC2018 dataset [44] is a publicly accessible dataset curated for dermatologic diagnosis by the International Skin Imaging Collaboration (ISIC). This dataset encompasses high-resolution images depicting various skin diseases, including malignant melanoma. Rigorous quality control and annotation processes have been applied to ensure the dataset's reliability. With over 10 000 images, here, we only adjust the format of the dataset to VOC format.

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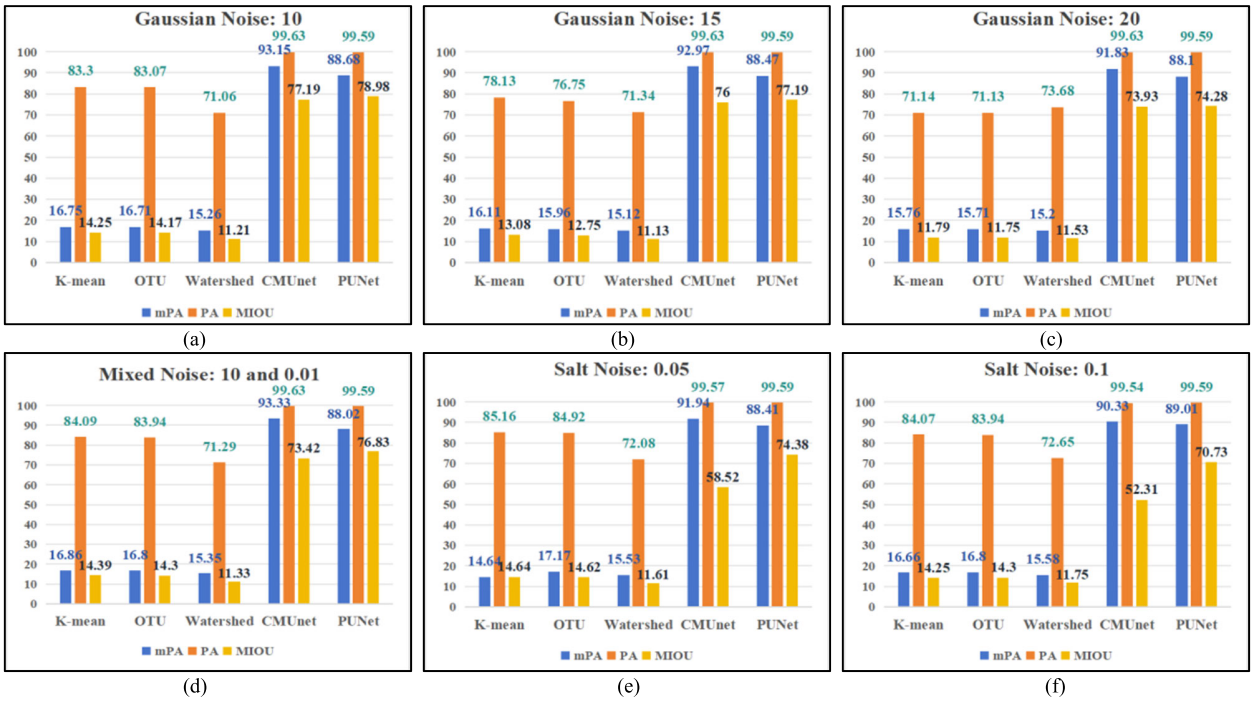


Fig. 5. Quantitative results of different kinds of noise. (a) Gaussian noise: 10. (b) Gaussian noise: 15. (c) Gaussian noise: 20. (d) Mixed noise: 10 and 0.01. (e) Salt noise: 0.05. (f) Salt noise: 0.1.

B. Comparison Experiment

To test the effectiveness of the P1 block enhancement, we initially introduced a convolution to the original C3 block, integrating it into the UNet architecture. We conducted comparative experiments on the PFDD. Specifically, we compared the performance of the modified C3 block with the P1 block within the UNet framework. The quantitative results of these experiments are detailed in Table III. Experiments demonstrate the effectiveness of our method. P1 block enhances the stability of each expanding convolutional branch, allowing it to adapt to channel variations more efficiently. In addition, the P1 module exhibits superior segmentation performance compared to the original C3 module under quantitative analysis of comparative experimentality.

The comparison experiment compares PUNet with other methods under different datasets to verify whether PUNet has advantages over other methods, whether it can be applied to other industrial datasets, and whether it has strong generalization ability.

We compared PUNet with other similar segmentation methods (DeeplabV3 [45], ResUNet [11], UNet++ [46], U-Net3+ [47], CMUNet [48], CMUeXt [49], HRNet [52], and MobileUNet [27]) on PFDD and TSDD. All methods are carried out in the same training environment and they were trained and validated using the same training set, validation set, and test set; Params and GFlops are measured at the same input size ($3 \times 256 \times 256$), and the quantitative metrics evaluation analysis is shown in Table IV, and the visualization effect is demonstrated in Fig. 4.

From Table IV, it can be seen that PUNet achieves excellent results compared to other methods with 14.27 M parametric quantities and 19.15 GFlops.

TABLE V
QUANTITATIVE ANALYSIS OF PUNET ON CRACK500

Method	PA(%)	CPA(%)	mIOU(%)
UNet[50]	97.09	88.07	80.57
UNet++[46]	**97.33**	86.90	81.25
UNet3+[47]	97.21	86.46	80.58
CMUNet[48]	97.28	87.89	81.36
CMUNext[49]	97.32	86.76	81.18
PUNet	96.42	**88.55**	**82.20**

PUNet performs better than other similar methods in terms of the shape of the defects, which achieves better segmentation effect than other similar methods with lower computational and parametric quantities, and has certain generalization ability in different industrial product surface defect datasets.

To validate the robustness performance of our method, we added different levels of Gaussian noise and salt-and-pepper noise to the images and conducted inference training on the noisy images using PUNet and other methods for comparative experimental analysis. The quantitative indicator analysis is shown in Fig. 5, and the visual result analysis is also presented in Fig. 6.

As shown in Figs. 5 and 6, we can see more intuitively that PUNet is less disturbed by noise than other methods, and can achieve defect location and segmentation in a more complete way under different circumstances.

To verify the segmentation performance of PUNet on Crack500, we compared it with UNet++ [46], UNet3+ [47], CMUNet [48], CMUeXt [49], and UNet [50] on the Crack500 dataset. The quantitative evaluation metrics are shown in Table V.

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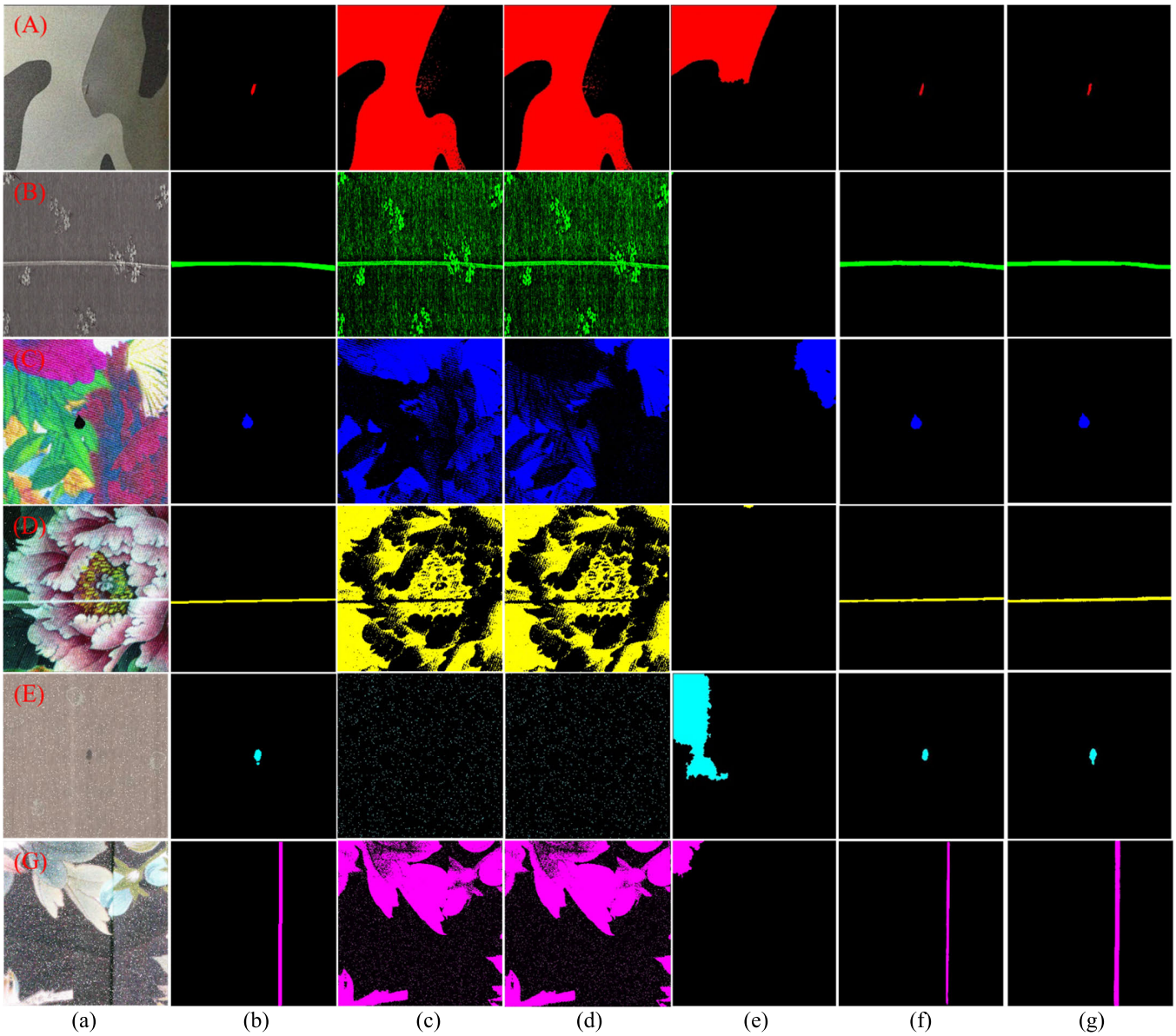


Fig. 6. Segmentation performance of different noises. (A) Gaussian noise: 10. (B) Gaussian noise: 15. (C) Gaussian noise: 20. (D) Mixed noise: 10 and 0.1. (E) Salt noise: 0.05. (F) Salt noise: 0.1. (a) Original images. (b) Label. (c) K-mean. (d) OUT. (e) Watershed. (f) CMUNet. (g) PUNet.

Collecting and annotating defect images of industrial products requires considerable cost, and most industrial product defect image datasets have fewer than 1000 images.

MVTEC dataset contains more categories of industrial products, each of which contains different kinds of defects, and some categories of industrial products have a limited number of samples. We select four of them (Leather, Pill, Srew, and Wood) of MVTEC for comparison to verify the effectiveness of PUNet on other industrial defects with a limited sample size. The evaluation of the quantitative indexes of the experiment is shown in Table VI, and the visualization result is shown in Fig. 7.

Fig. 7 shows that PUNet has generalization ability, which is not only applicable to printed fabric defects but also has excellent performance on other industrial defect datasets.

We compared PUNet with other methods at ISIC2018. The experimental configurations are the same as described above,

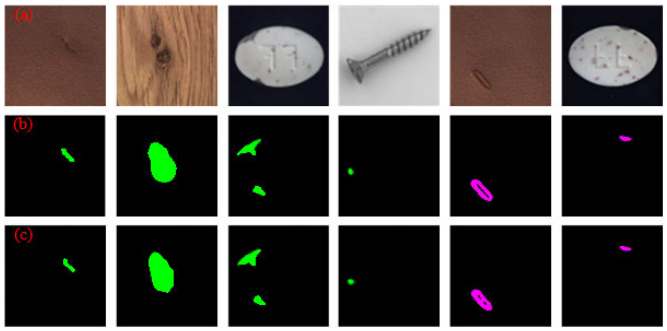


Fig. 7. Segmentation performance of PUNet on the MVTEC dataset. (a) Original images. (b) Labels. (c) PUNet.

and we obtained quantitative metrics evaluated as shown in Table VII.

From Table VII, it can be seen that PUNet achieves excellent segmentation performance on the ISIC2018 dataset with

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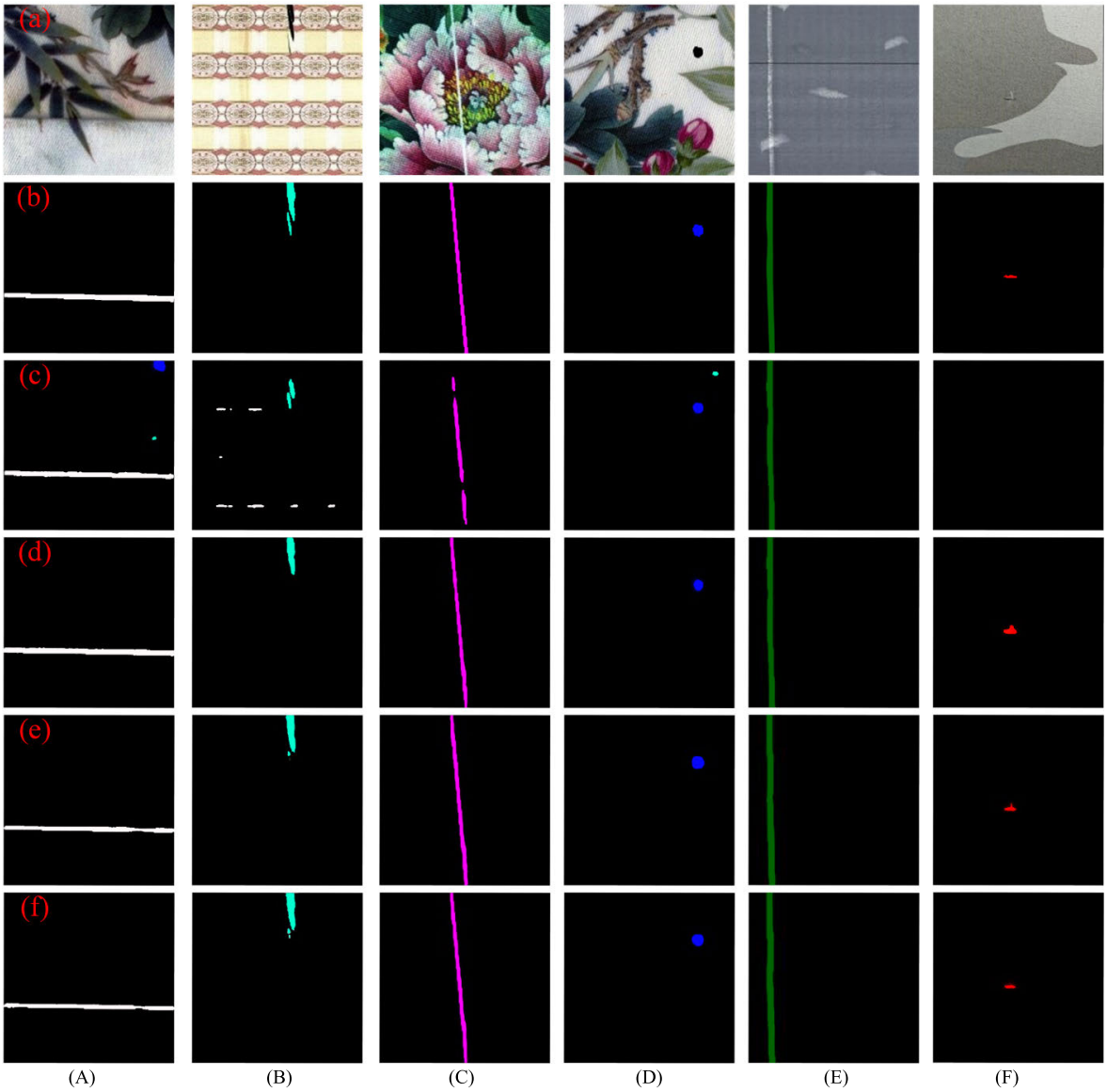


Fig. 8. Segmentation performance of PUNet and UNet on printed fabric defect dataset. (a) Original images. (b) Labels. (c) UNet. (d) UNet + P1 block. (e) UNet + PSA. (f) PUNet. (A) Wrinkles (Wi). (B) Stains (St). (C) Pass (Pa).

TABLE VI
QUANTITATIVE ANALYSIS OF PUNET ON MVTEC DATASET

Dataset		UNet [50]	DeepLabV 3+[45]	CMUNet [48]	PUNet
Leather	PA(%)	99.70	97.63	99.75	**99.71**
	mPA(%)	79.11	79.14	86.43	**91.68**
	mIOU(%)	73.09	74.89	**77.94**	77.93
Pill	PA(%)	98.34	96.57	98.20	**99.45**
	mPA(%)	48.71	22.75	73.10	**84.23**
	mIOU(%)	36.48	14.53	52.52	**72.48**
Screw	PA(%)	99.85	98.68	99.84	**99.85**
	mPA(%)	75.52	76.53	**79.58**	78.51
	mIOU(%)	67.15	64.25	**71.32**	68.35
Wood	PA(%)	97.93	97.50	98.16	**98.41**
	mPA(%)	88.25	72.08	86.57	**88.46**
	mIOU(%)	67.96	58.62	72.04	**88.86**

a 1.93% improvement in the mIOU of UNet. The experiments demonstrate that PUNet has excellent segmentation performance and generalization ability, and outperforms most existing similar methods on the ISIC2018.

C. Ablation Experiment

In this section, two experiments will be conducted to validate the efficacy of PUNet for defect detection of printed fabrics and to assess its performance under different backbones.

In the printed fabric defect dataset, numerous defects exhibit complex backgrounds and varying sizes, posing challenges for pixel-level defect segmentation. As depicted in Fig. 8, the segmentation performance of UNet on the printing defects dataset exists in mis-segmentation and the potential loss of small defects.

PUNet incorporates the P1 block to mitigate the loss of small defective targets within complex backgrounds, mitigate mis-segmentation resulting from complex backgrounds, and leverage the PSA module to enhance the segmentation of edge

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TABLE VII
QUANTITATIVE ANALYSIS OF PUNET ON THE ISIC2018 DATASET

Method	Acc(%)	SE(%)	SP(%)	mIOU(%)
UNet[50]	95.39	87.35	97.44	81.01
ResUNet++	93.82	87.35	97.21	81.00
Atten-SwinUNet[53]	94.80	80.57	98.26	-
LightM-UNet[54]	95.10	88.29	**97.65**	-
Ultra-Light-VMUNet[55]	95.58	86.80	97.81	-
PUNet	**96.15**	**90.10**	**97.65**	**83.11**

TABLE VIII
QUANTITATIVE EVALUATION OF PSA AND P1 BLOCK ON PRINTED FABRIC DEFECT DATASET

Method/Class				HF	BW	IL	Pa	St	Wr
UNet	P1	PSA							
√			CPA	0.42	0.90	0.97	0.89	0.45	0.66
√	√		CPA	0.67	0.92	**0.98**	0.94	0.64	**0.97**
√		√	CPA	**0.80**	0.93	**0.98**	0.82	0.63	0.92
√	√	√	CPA	**0.80**	**0.94**	**0.98**	**0.95**	**0.66**	0.91
√			IOU	0.41	**0.84**	**0.96**	0.82	0.45	0.66
√	√		IOU	0.56	**0.84**	**0.96**	**0.86**	0.58	0.72
√		√	IOU	**0.64**	**0.84**	**0.96**	0.80	0.57	0.74
√	√	√	IOU	**0.64**	**0.84**	0.95	**0.86**	**0.62**	**0.75**

TABLE IX
QUANTITATIVE EVALUATION OF PSA AND P1 BLOCK WITH DIFFERENT BACKBONES

Method				mPA	mIOU	Params	GFlops
UNet	P1	PSA	Backbone				
			Conv	77.20	73.50	31.04	46.15
√			Resnet50	84.59	76.16	43.93	22.94
			VGG16	87.50	82.30	24.89	56.38
			Conv	87.36	78.88	11.47	**12.67**
√	√		Resnet50	84.63	80.11	**24.72**	**11.85**
			VGG16	89.07	82.74	**16.58**	**26.11**
			Conv	86.69	79.47	33.84	52.76
√		√	Resnet50	88.20	81.15	65.53	30.72
			VGG16	89.26	**82.94**	28.55	68.92
			Conv	**89.23**	**80.66**	14.27	19.15
√	√	√	Resnet50	**88.64**	**82.54**	47.32	19.63
			VGG16	**89.50**	82.61	20.24	38.65

details for large defective targets. To validate the effectiveness of PUNet, we assess the individual module and the combined modules in comparison to the UNet. The quantitative indices are evaluated and presented in Table VIII, and the visual comparison is depicted in Fig. 8.

The experimental results show that our method is effective, the P1 block improves the network's ability to perceive defects under complex backgrounds, but the difference caused by the P1 block in the channel dimension causes the model to ignore the defect detail information, such as the class defects of Wrinkles, Pass, and Stain, whereas the addition of the PSA module suppresses this difference and enhances the network's extraction of the detail information of the defect edges.

To verify whether PUNet is still effective under different backbones, we conduct ablation experiments under the original backbone, Resnet-50, and VGG-16 three different backbones, and the quantitative metrics are evaluated as shown in Table IX.

PUNet reduces 16.77 M parameters and 27 GFlops, which improves mIOU and mPA by 7.16% and 12.03%, respectively, compared to UNet with the backbone as convolution. With a backbone of VGG16 or Resnet-50, the mAP and mIOU boosts, as well as the decreases in the number of parameters and computation, are still valid.

The experimental results show that PUNet can be effectively applied to the complex diversity of defects in printed fabrics, and the segmentation effect of PUNet for defect targets of different sizes is significantly improved compared with that of UNet.

V. CONCLUSION

In this article, PUNet has been proposed for the defect segmentation of printed fabrics. Extensive experiments on various types of datasets show the capability of our method, which can efficiently achieve accurate segmentation of printed fabric defects with different sizes, excellent generalization ability, and stronger segmentation effect achieved with very low computational cost compared to UNet. The comparisons between our method and the state-of-the-art methods demonstrate that our proposal outperforms existing schemes in terms of both qualitative and quantitative. In future work, we will try to extend our proposal to reduce the demand for labeled data to realize the accurate segmentation of the printed fabric defects.

REFERENCES

[1] C. Kumah, R. K. Raji, and R. Pan, "Review of printed fabric pattern segmentation analysis and application," *Autex Res. J.*, vol. 20, no. 4, pp. 530–538, Dec. 2020.

[2] Y. Gao, X. Li, X. V. Wang, L. Wang, and L. Gao, "A review on recent advances in vision-based defect recognition towards industrial intelligence," *J. Manuf. Syst.*, vol. 62, pp. 753–766, Jan. 2022.

[3] P. M. Bhatt et al., "Image-based surface defect detection using deep learning: A review," *J. Comput. Inf. Sci. Eng.*, vol. 21, no. 4, Aug. 2021, Art. no. 040801.

[4] J. Yang, S. Li, Z. Wang, H. Dong, J. Wang, and S. Tang, "Using deep learning to detect defects in manufacturing: A comprehensive survey and current challenges," *Materials*, vol. 13, no. 24, p. 5755, Dec. 2020.

[5] X. Tao, D. Zhang, W. Ma, X. Liu, and D. Xu, "Automatic metallic surface defect detection and recognition with convolutional neural networks," *Appl. Sci.*, vol. 8, no. 9, p. 1575, Sep. 2018.

[6] J. Jing and H. Ren, "Defect detection of printed fabric based on RGBAAM and image pyramid," *Autex Res. J.*, vol. 21, no. 2, pp. 135–141, Jun. 2021.

[7] A. Liu, E. Yang, J. Wu, Y. Teng, and L. Yu, "Double sparse low rank decomposition for irregular printed fabric defect detection," *Neurocomputing*, vol. 482, pp. 287–297, Apr. 2022.

[8] Z. Pourkaramdel, S. Fekri-Ershad, and L. Nanni, "Fabric defect detection based on completed local quartet patterns and majority decision algorithm," *Expert Syst. Appl.*, vol. 198, Jul. 2022, Art. no. 116827.

[9] B. Wan et al., "LFRNet: Localizing, focus, and refinement network for salient object detection of surface defects," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.

[10] N. Zeng, P. Wu, Z. Wang, H. Li, W. Liu, and X. Liu, "A small-sized object detection oriented multi-scale feature fusion approach with application to defect detection," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–14, 2022.

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[11] X. Xiao, S. Lian, Z. Luo, and S. Li, "Weighted res-UNet for high-quality retina vessel segmentation," in *Proc. 9th Int. Conf. Inf. Technol. Med. Educ. (ITME)*, Oct. 2018, pp. 327–331.

[12] S. Guan, A. A. Khan, S. Sikdar, and P. V. Chitnis, "Fully dense UNet for 2-D sparse photoacoustic tomography artifact removal," *IEEE J. Biomed. Health Informat.*, vol. 24, no. 2, pp. 568–576, Feb. 2020.

[13] A. Lou, S. Guan, and M. H. Loew, "DC-UNet: Rethinking the U-Net architecture with dual channel efficient CNN for medical image segmentation," in *Medical Imaging 2021: Image Processing*. Bellingham, WA, USA: SPIE, 2021.

[14] N. Ibtehaz and M. S. Rahman, "MultiResUNet : Rethinking the U-Net architecture for multimodal biomedical image segmentation," *Neural Netw.*, vol. 121, pp. 74–87, Jan. 2020.

[15] J. M. J. Valanarasu, V. A. Sindagi, I. Hacihaliloglu, and V. M. Patel, "KiU-net: Overcomplete convolutional architectures for biomedical image and volumetric segmentation," *IEEE Trans. Med. Imag.*, vol. 41, no. 4, pp. 965–976, Apr. 2022.

[16] P. Ahmad et al., "MH UNet: A multi-scale hierarchical based architecture for medical image segmentation," *IEEE Access*, vol. 9, pp. 148384–148408, 2021.

[17] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[18] J. Guo, H.-Y. Zhou, L. Wang, and Y. Yu, "UNet-2022: Exploring dynamics in non-isomorphic architecture," 2022, *arXiv:2210.15566*.

[19] A. Lin, B. Chen, J. Xu, Z. Zhang, G. Lu, and D. Zhang, "DS-TransUNet: Dual Swin transformer U-Net for medical image segmentation," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–15, 2022.

[20] L. Cheng, J. Yi, A. Chen, and Y. Zhang, "Fabric defect detection based on separate convolutional UNet," *Multimedia Tools Appl.*, vol. 82, no. 2, pp. 3101–3122, Jan. 2023.

[21] Z. Lei, J. Junfeng, and Z. Hongwei, "Fabric defect classification based on LBP and GLCM," *J. Fiber Bioeng. Inform.*, vol. 8, no. 1, pp. 81–89, 2015.

[22] J. Jing, M. Zheng, H. Zhang, and D. Zhuo, "A coarse-to-fine method for glass fiber fabric surface defect detection," *J. Textile Inst.*, vol. 112, no. 3, pp. 388–397, 2021.

[23] K. Zhang et al., "Cross-scale cost aggregation for stereo matching," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1590–1597.

[24] D.-M. Tsai and C.-K. Huang, "Defect detection in electronic surfaces using template-based Fourier image reconstruction," *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 9, no. 1, pp. 163–172, Jan. 2019.

[25] X. Kang and E. Zhang, "A universal defect detection approach for various types of fabrics based on the elo-rating algorithm of the integral image," *Textile Res. J.*, vol. 89, nos. 21–22, pp. 4766–4793, Nov. 2019.

[26] H. Zhang, J. Ma, J. Jing, and P. Li, "Fabric defect detection using L0 gradient minimization and fuzzy C-means," *Appl. Sci.*, vol. 9, no. 17, p. 3506, Aug. 2019.

[27] J. Jing, Z. Wang, M. Rätsch, and H. Zhang, "Mobile-Unet: An efficient convolutional neural network for fabric defect detection," *Textile Res. J.*, vol. 92, nos. 1–2, pp. 30–42, Jan. 2022.

[28] Y. Huang, J. Jing, and Z. Wang, "Fabric defect segmentation method based on deep learning," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–15, 2021.

[29] H. Zhang, H. Zhu, J. Jing, P. Li, and Q. Pan, "Curve-like structure detection using multiscale and boundary-assisted segmentation network," *IEEE Trans. Instrum. Meas.*, vol. 73, 2024, Art. no. 5007215.

[30] R. Ozdemir and M. Koc, "On the enhancement of semi-supervised deep learning-based railway defect detection using pseudo-labels," *Expert Syst. Appl.*, vol. 251, Oct. 2024, Art. no. 124105.

[31] Y. Liu, H. Yang, and C. Wu, "Unveiling patterns: A study on semi-supervised classification of strip surface defects," *IEEE Access*, vol. 11, pp. 119933–119946, 2023.

[32] M. Yang, P. Wu, and H. Feng, "MemSeg: A semi-supervised method for image surface defect detection using differences and commonalities," *Eng. Appl. Artif. Intell.*, vol. 119, Mar. 2023, Art. no. 105835.

[33] X. Hu, J. Yang, F. Jiang, A. Hussain, K. Dashtipour, and M. Gogate, "Steel surface defect detection based on self-supervised contrastive representation learning with matching metric," *Appl. Soft Comput.*, vol. 145, Sep. 2023, Art. no. 110578.

[34] H. Zhang, W. Xiong, S. Lu, M. Chen, and L. Yao, "QA-USTNet: Yarn-dyed fabric defect detection via U-shaped Swin transformer network based on quadtree attention," *Textile Res. J.*, vol. 93, nos. 15–16, pp. 3492–3508, Aug. 2023.

[35] H.-W. Zhang, Q.-L. Tan, S. Lu, Z.-Q. Ge, and D. Gu, "Yarn-dyed fabric defect detection using U-shaped de-noising convolutional auto-encoder," in *Proc. IEEE 9th Data Driven Control Learn. Syst. Conf. (DDCLS)*, Nov. 2020, pp. 18–24.

[36] H. Zhang, G. Qiao, S. Lu, L. Yao, and X. Chen, "Attention-based feature fusion generative adversarial network for yarn-dyed fabric defect detection," *Textile Res. J.*, vol. 93, nos. 5–6, pp. 1178–1195, Mar. 2023.

[37] T. Zhang and H. Ma, "ClothNet: Sensitive semantic segmentation network for fabric defect detection," *Textile Res. J.*, vol. 93, nos. 1–2, pp. 103–115, Jan. 2023.

[38] H. Zhang, S. Wang, S. Lu, L. Yao, and Y. Hu, "Attention-gate-based U-shaped reconstruction network (AGUR-Net) for color-patterned fabric defect detection," *Textile Res. J.*, vol. 93, nos. 15–16, pp. 3459–3477, Aug. 2023.

[39] G. Hu, J. Huang, Q. Wang, J. Li, Z. Xu, and X. Huang, "Unsupervised fabric defect detection based on a deep convolutional generative adversarial network," *Textile Res. J.*, vol. 90, nos. 3–4, pp. 247–270, 2020.

[40] H. Park, Y. Yoo, G. Seo, D. Han, S. Yun, and N. Kwak, "C3: Concentrated-comprehensive convolution and its application to semantic segmentation," 2018, *arXiv:1812.04920*.

[41] Y. Huang, C. Qiu, and K. Yuan, "Surface defect saliency of magnetic tile," *Vis. Comput.*, vol. 36, no. 1, pp. 85–96, Jan. 2020.

[42] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, "MVTec AD—A comprehensive real-world dataset for unsupervised anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 9592–9600.

[43] F. Yang, L. Zhang, S. Yu, D. Prokhorov, X. Mei, and H. Ling, "Feature pyramid and hierarchical boosting network for pavement crack detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 4, pp. 1525–1535, Apr. 2020.

[44] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," *Sci. Data*, vol. 5, no. 1, pp. 1–9, Aug. 2018.

[45] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[46] Z. Zhou, M. M. Rahman Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-net architecture for medical image segmentation," in *Proc. Deep Learn. Med. Image Anal. Multimodal Learn. Clin. Decis. Support: 4th Int. Workshop, DLMIA, 8th Int. Workshop, ML-CDS, Held Conjunct (MICCAI)*, Granada, Spain. Heidelberg, Germany: Springer, pp. 3–11.

[47] H. Huang et al., "UNet 3+: A full-scale connected UNet for medical image segmentation," 2020, *arXiv:2004.08790*.

[48] F. Tang, L. Wang, C. Ning, M. Xian, and J. Ding, "CMU-Net: A strong convmixer-based medical ultrasound image segmentation network," in *Proc. IEEE 20th Int. Symp. Biomed. Imag. (ISBI)*, Apr. 2023, pp. 1–5.

[49] F. Tang, J. Ding, L. Wang, C. Ning, and S. Kevin Zhou, "CMUNeXt: An efficient medical image segmentation network based on large kernel and skip fusion," 2023, *arXiv:2308.01239*.

[50] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*. Munich, Germany: Springer, pp. 234–241.

[51] L. C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 801–818.

[52] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5693–5703.

[53] E. K. Aghdam, R. Azad, M. Zarvani, and D. Merhof, "Attention Swin U-net: Cross-contextual attention mechanism for skin lesion segmentation," in *Proc. IEEE 20th Int. Symp. Biomed. Imag. (ISBI)*, Apr. 2023, pp. 1–5.

[54] W. Liao, Y. Zhu, X. Wang, C. Pan, Y. Wang, and L. Ma, "LightM-UNet: Mamba assists in lightweight UNet for medical image segmentation," 2024, *arXiv:2403.05246*.

[55] R. Wu, Y. Liu, P. Liang, and Q. Chang, "UltraLight VM-UNet: Parallel vision mamba significantly reduces parameters for skin lesion segmentation," 2024, *arXiv:2403.20035*.

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Jie Tang received the B.S. degree from Xi'an Polytechnic University, Xi'an, China, in 2021, where he is currently pursuing the master's degree.

His current research interests include deep learning and image segmentation.



Meng Jiang is currently a Teacher at Xi'an Polytechnic University, Xi'an, China. Her main research interests include artificial intelligence and deep learning.



Huanhuan Zhang received the Ph.D. degree from Xi'an University of Technology, Xi'an, China, in 2020.

She is an Associate Professor at Xi'an Polytechnic University, Xi'an. Her current research interests include deep learning, computer vision, and image processing.



Pengfei Li is currently a Professor at Xi'an Polytechnic University, Xi'an, China. His main research interests include machine learning, data mining, and image processing.



Zebin Su is currently a Senior Engineer at Xi'an Polytechnic University, Xi'an, China. His main research interests include artificial intelligence, machine learning, and image generation.



Junfeng Jing received the M.S. degree in electrical engineering from Xi'an Polytechnic University, Xi'an, China, in 2006, and the Ph.D. degree in mechatronic engineering from Xidian University, Xi'an, in 2013.

He is a Professor at Xi'an Polytechnic University, Xi'an. His research interests include artificial intelligence, machine vision, and industrial robot control.

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